

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key. **100 points:** Part A 20, Part B 20, Part C 40, Part D 20.

Part A — Vocabulary

10 questions, 2 points each

Write the letter of the matching description next to each term.

- | | |
|--------------------------------|--|
| 1. _____ Population | A. A number that describes a sample. |
| 2. _____ Sample | B. The researcher assigns people to groups and gives them a treatment. |
| 3. _____ Parameter | C. A variable whose values are labels, like a color or a team name. |
| 4. _____ Statistic | D. The whole group we want to learn about. |
| 5. _____ Categorical variable | E. Everyone has an equal chance of being picked. |
| 6. _____ Quantitative variable | F. The researcher watches or asks, without assigning anyone to a group. |
| 7. _____ Simple random sample | G. A number that describes a whole population. |
| 8. _____ Convenience sample | H. The part of the population we actually collect data from. |
| 9. _____ Observational study | I. A variable whose values are measured amounts, like minutes or inches. |
| 10. _____ Experiment | J. Choosing whoever is easiest to reach. |

Part B — Multiple Choice

5 questions, 4 points each

11. Which of these variables is **quantitative**?
 - (A) Eye color
 - (B) Favorite sport
 - (C) Height in inches
 - (D) Home ZIP code
12. Maplewood High School has 900 students. A survey of 60 of them finds that 30% walk to school. The 30% is a
 - (A) statistic, because it describes the sample.
 - (B) parameter, because it describes the sample.
 - (C) statistic, because it describes all 900 students.
 - (D) parameter, because it describes all 900 students.
13. A librarian takes an alphabetical list of card holders, picks a random start, and then chooses **every 10th name**. This is a
 - (A) stratified sample.
 - (B) cluster sample.
 - (C) convenience sample.
 - (D) systematic sample.

14. To learn how often town residents exercise, a researcher surveys people **inside a gym**. The results will most likely
 - (A) be about right, because the people were chosen fairly.
 - (B) overestimate how often residents exercise.
 - (C) underestimate how often residents exercise.
 - (D) be right if the researcher surveys more people at the gym.
15. Researchers use a coin flip to put 80 volunteers into two groups. One group sleeps 8 hours and the other sleeps 5 hours before a memory test. This study is
 - (A) an experiment, because the researchers assigned the groups.
 - (B) an experiment, because it measures memory.
 - (C) an observational study, because the volunteers were watched.
 - (D) an observational study, because there are two groups.

Part C — Short Computations

4 questions, 10 points each

16. **Silver Lake Recreation Center** has 900 members. The director asks 60 members chosen at random, and 18 of them use the pool every week.
 - (a) What percent of the sample uses the pool every week?
 - (b) About how many of all 900 members use the pool every week?
17. **Granite Falls Middle School** has 600 students and wants a **stratified** sample of 60 — the same fraction from each grade.
 - (a) What fraction of the school is being sampled?

(b) Complete the table.

Grade	6	7	8	Total
Students in grade	240	200	160	600
Number sampled				

18. **Brookside Apartments** has 500 households on a numbered list. The manager wants a **systematic** sample of 50.
- (a) Find the interval k .

(b) The random start is household 3. List the first four households chosen.

19. In an **experiment**, 40 students are randomly assigned to use a flashcard app and 40 to study the usual way. Afterwards, 30 of the app group and 20 of the usual group passed a quiz.
- (a) Find the percent who passed in each group.

(b) How many percentage points apart are the two groups? _____

Part D — Critical Thinking

1 question, 20 points

20. A random sample of Maplewood students found that students who **eat breakfast** have higher grades than students who skip it. Nobody was assigned — students chose for themselves. The principal says, “*Eating breakfast causes higher grades.*”

Is the principal’s conclusion justified? Explain your reasoning, and name one **other variable** that could explain the result.
